TASK 5: Resize the Disk of a Windows Virtual Machine

Objective

To increase the size of an Azure Managed Data Disk attached to a Windows Virtual Machine and extend the volume inside Windows to utilize the additional storage space.

Theory

Azure Disk Resizing allows administrators to increase the capacity of an existing managed disk without recreating the virtual machine. After increasing the disk size from the Azure Portal, the additional storage appears as unallocated space inside the operating system and must be extended using Disk Management.

Benefits

- Increases disk capacity without data loss.
- Eliminates the need to create a new disk.
- Supports growing storage requirements.
- Simple and quick administration process.

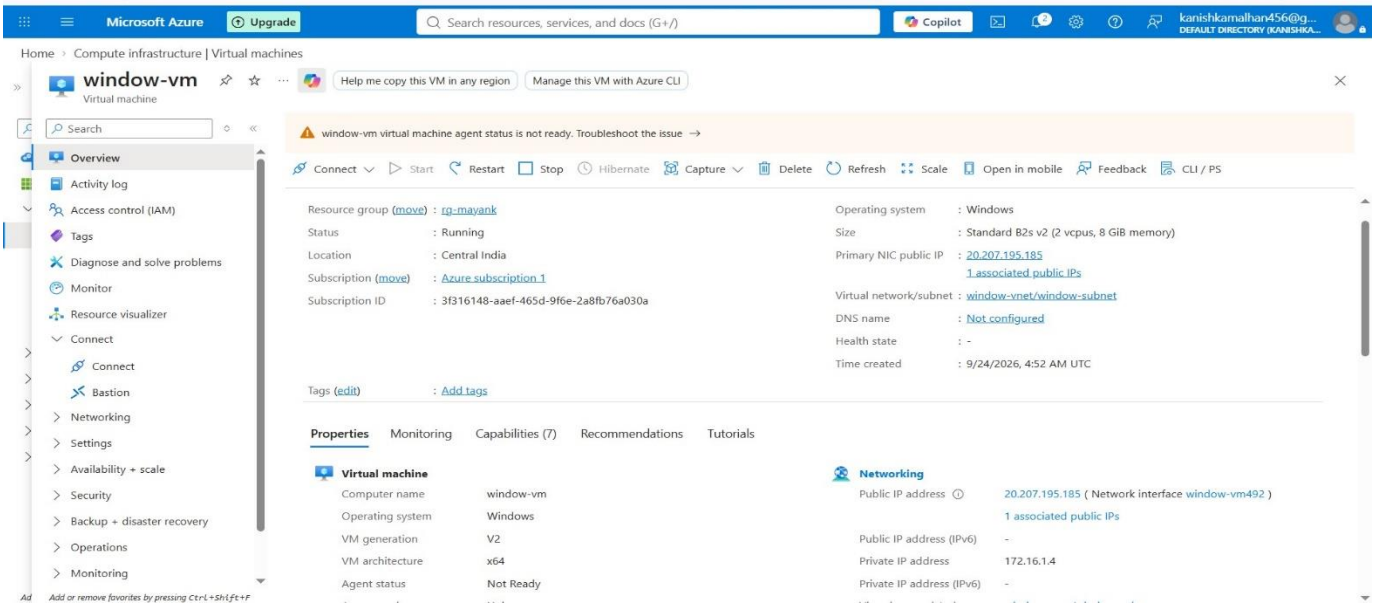
Components Used

- Microsoft Azure Portal
- Windows Virtual Machine (window-vm)
- Azure Managed Disk
- Windows Disk Management

Procedure

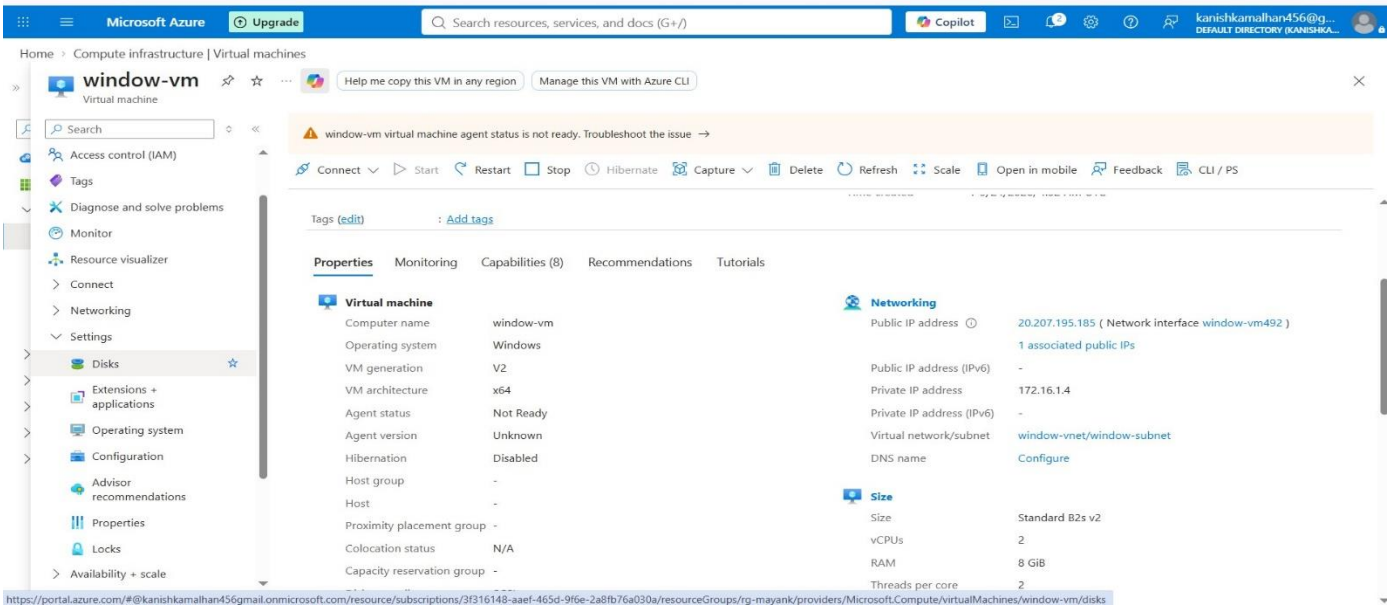
Step 1: Open the Windows Virtual Machine

1. Sign in to the Azure Portal.
2. Navigate to **Virtual Machines**.
3. Open the VM named **window-vm**.
4. Verify that the VM is in the Running state.



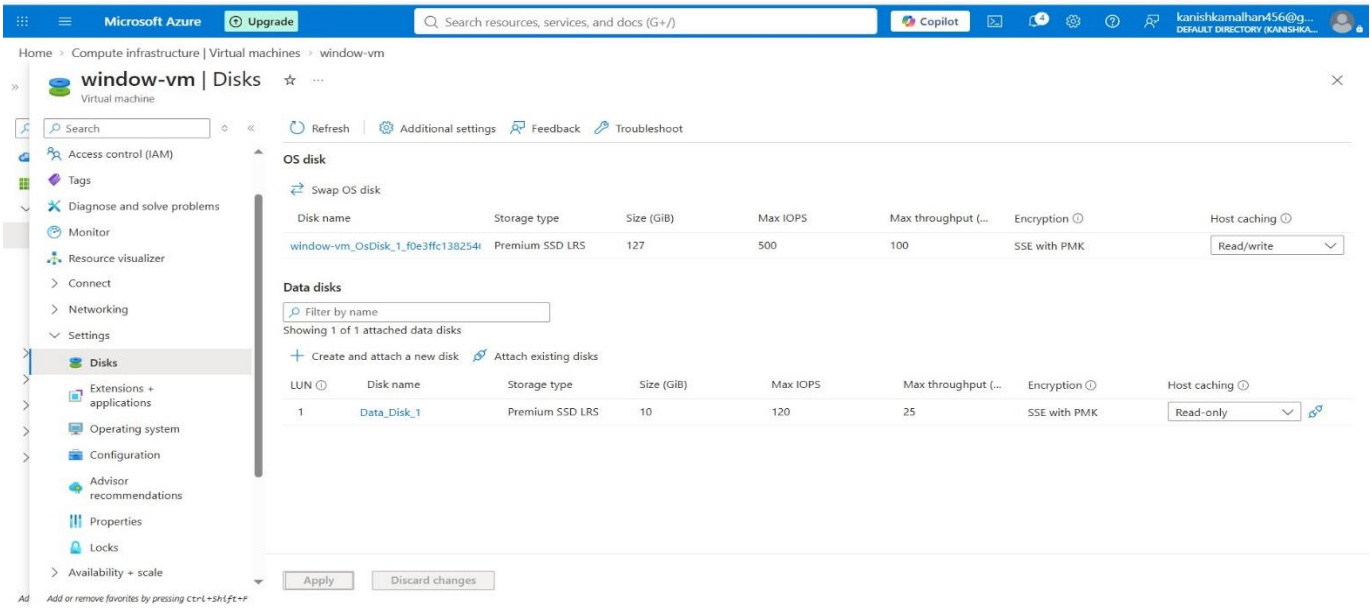
Step 2: Open Disk Settings

1. In the VM's left navigation pane, click **Disks** under Settings.
2. View the attached OS Disk and Data Disk.



Step 3: Verify Existing Data Disk Configuration

1. Open the VM Disks page.
2. Review the attached data disk details.
3. Verify that **Data_Disk1** is currently attached.

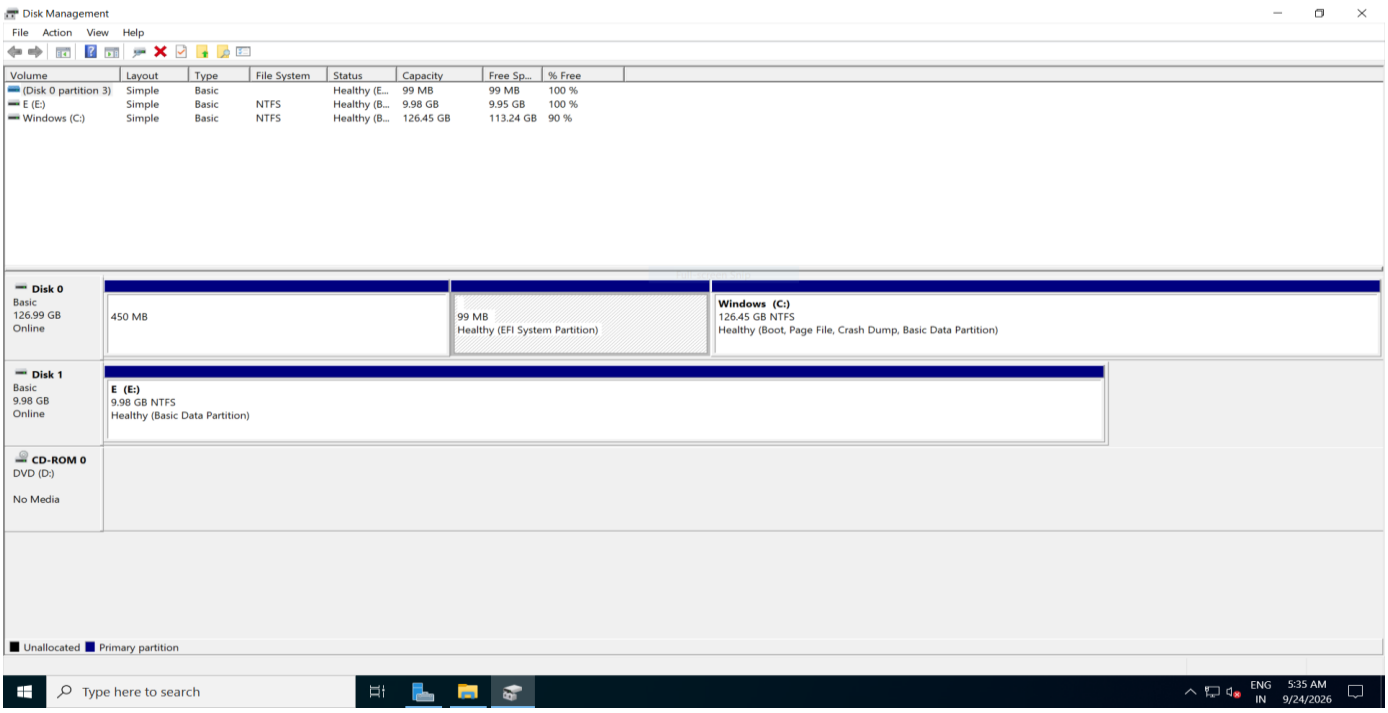


Step 4: Open Data Disk Overview

1. Click on **Data_Disk1**.
2. Review disk properties and current size.

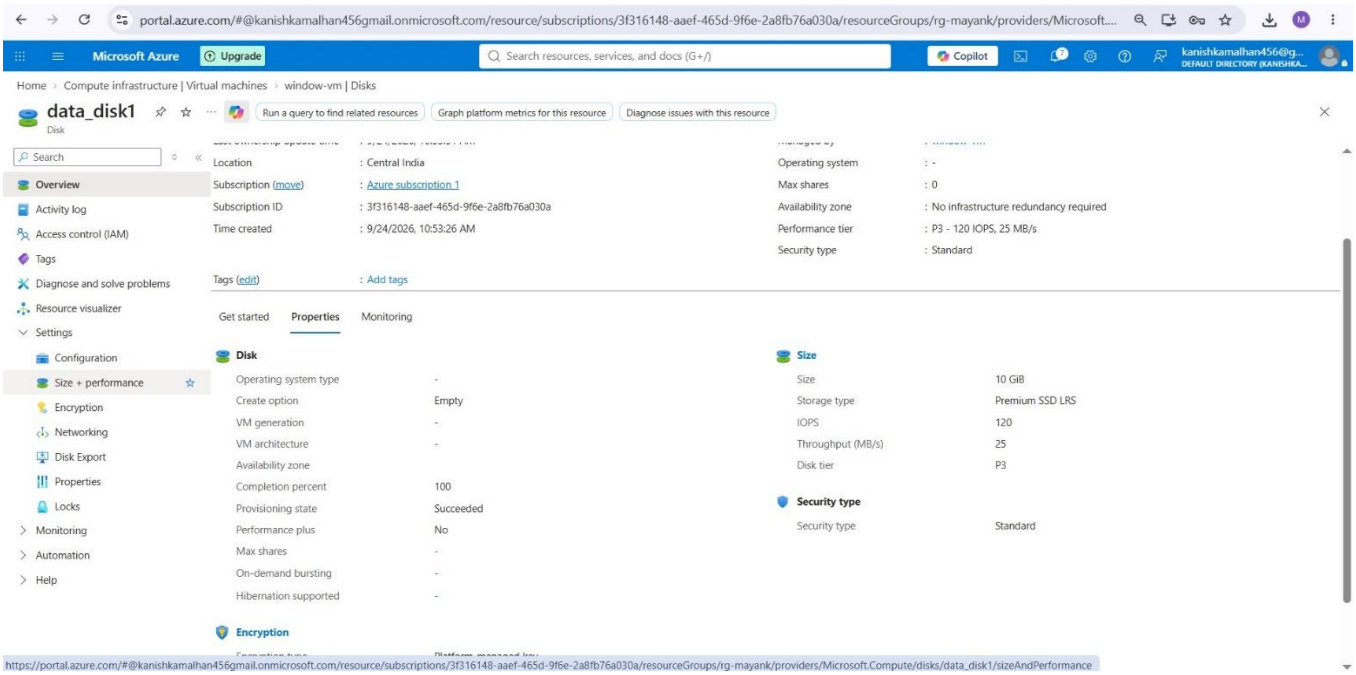
Existing Disk Details

- Disk Name: data_disk1
- Storage Type: Premium SSD LRS
- Current Size: 10 GiB
- IOPS: 120
- Disk Tier: P3



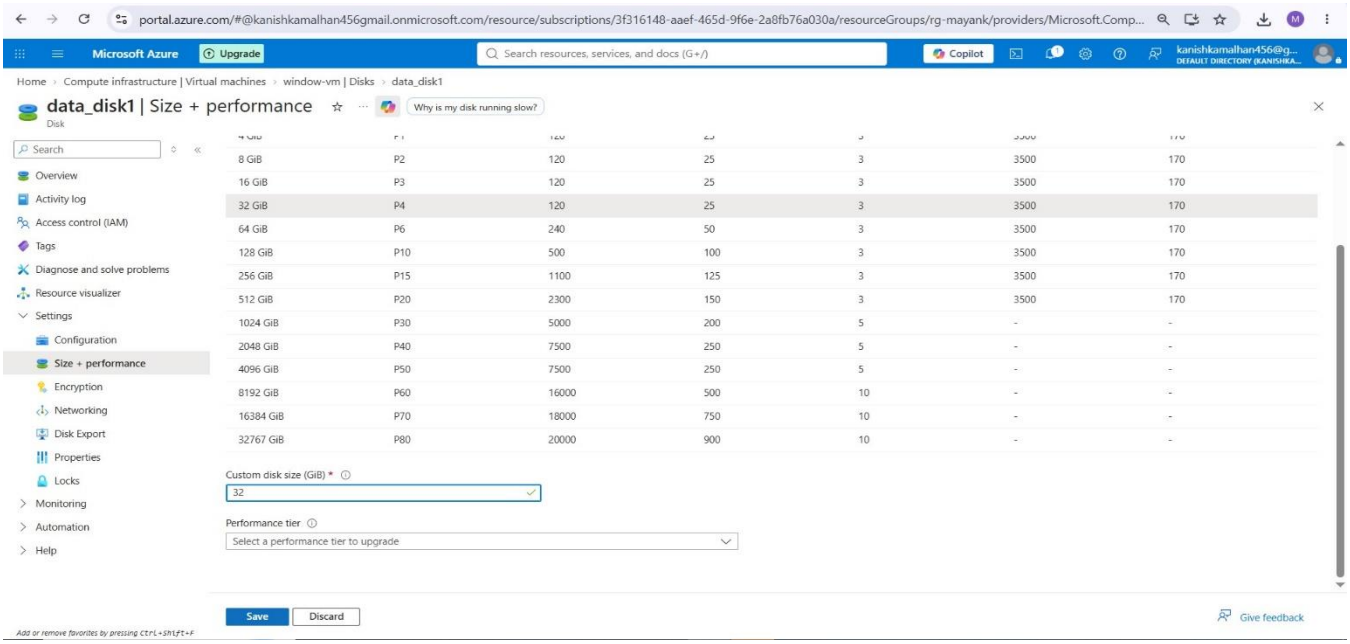
Step 5: Open Size + Performance Settings

1. From the left menu, select **Size + Performance**.
2. Review available disk size options.



Step 6: Increase Disk Size

1. In the **Custom Disk Size (GiB)** field, enter **32**.
2. Change the disk size from **10 GiB** to **32 GiB**.
3. Click **Save**.

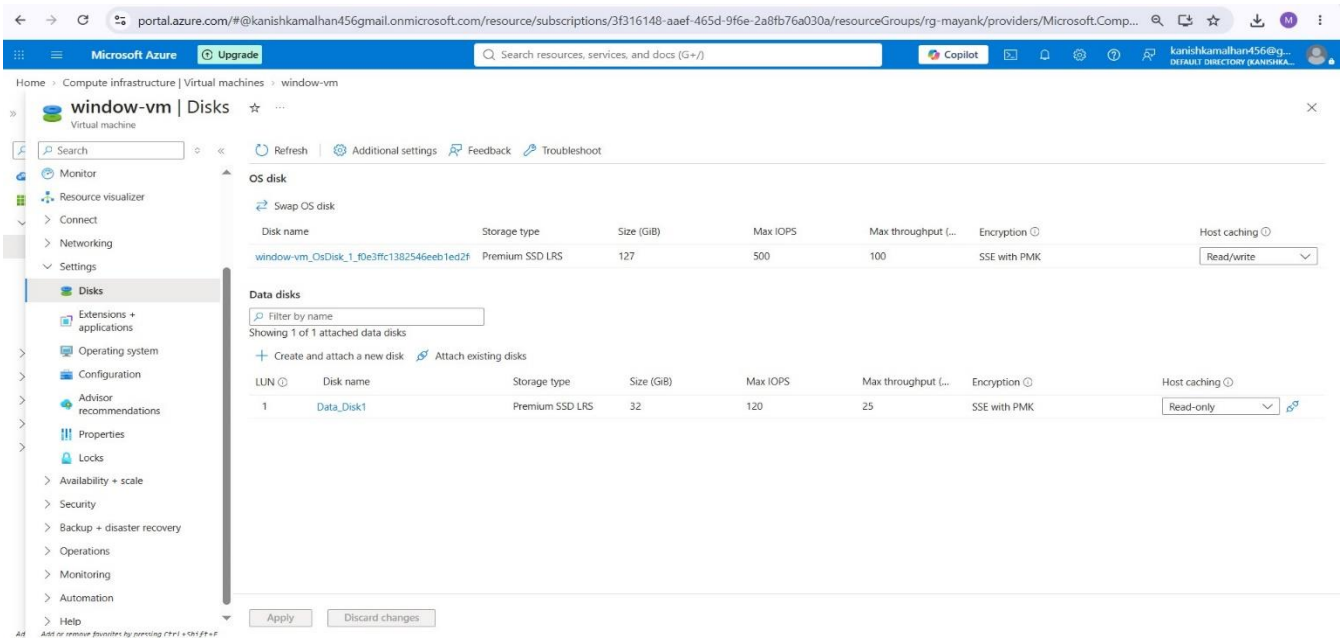


Step 7: Verify Disk Resize in Azure Portal

1. Return to the VM Disks page.
2. Verify that the size of **Data_Disk1** has been updated to **32 GiB**.

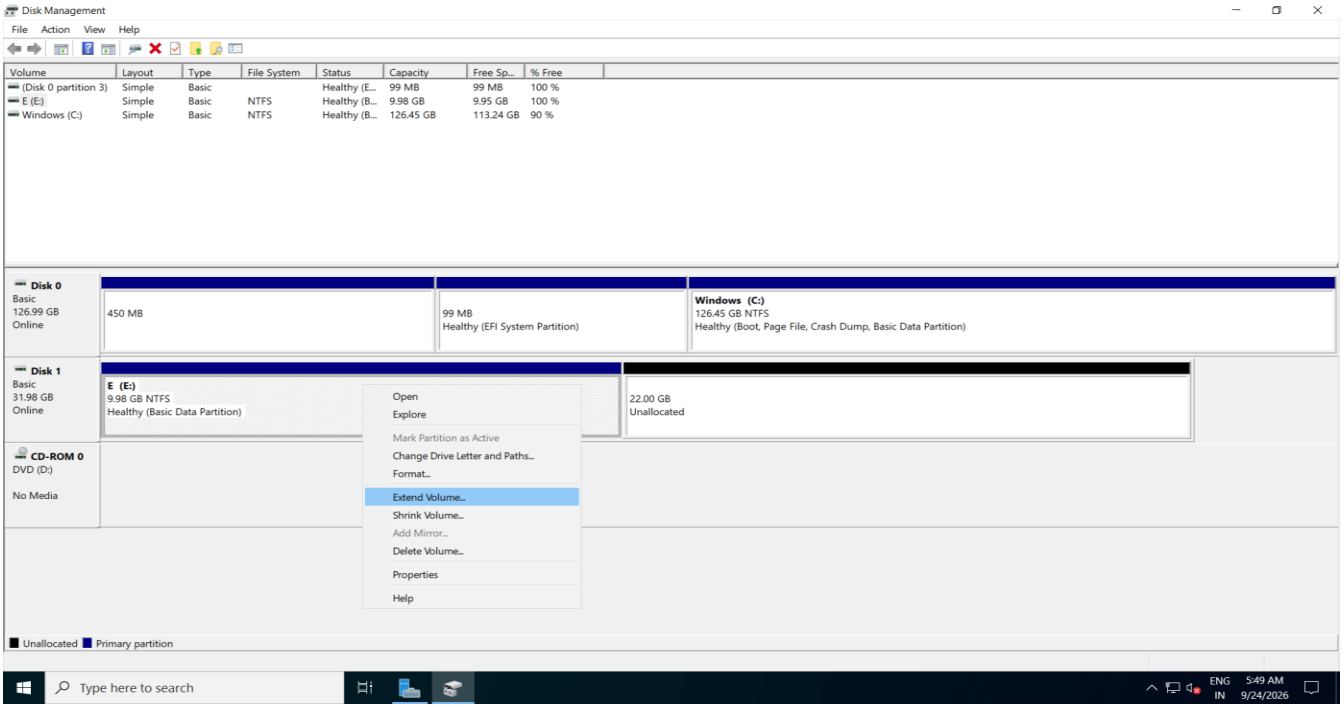
Updated Disk Details

- Disk Name: Data_Disk1
- Storage Type: Premium SSD LRS
- Updated Size: 32 GiB



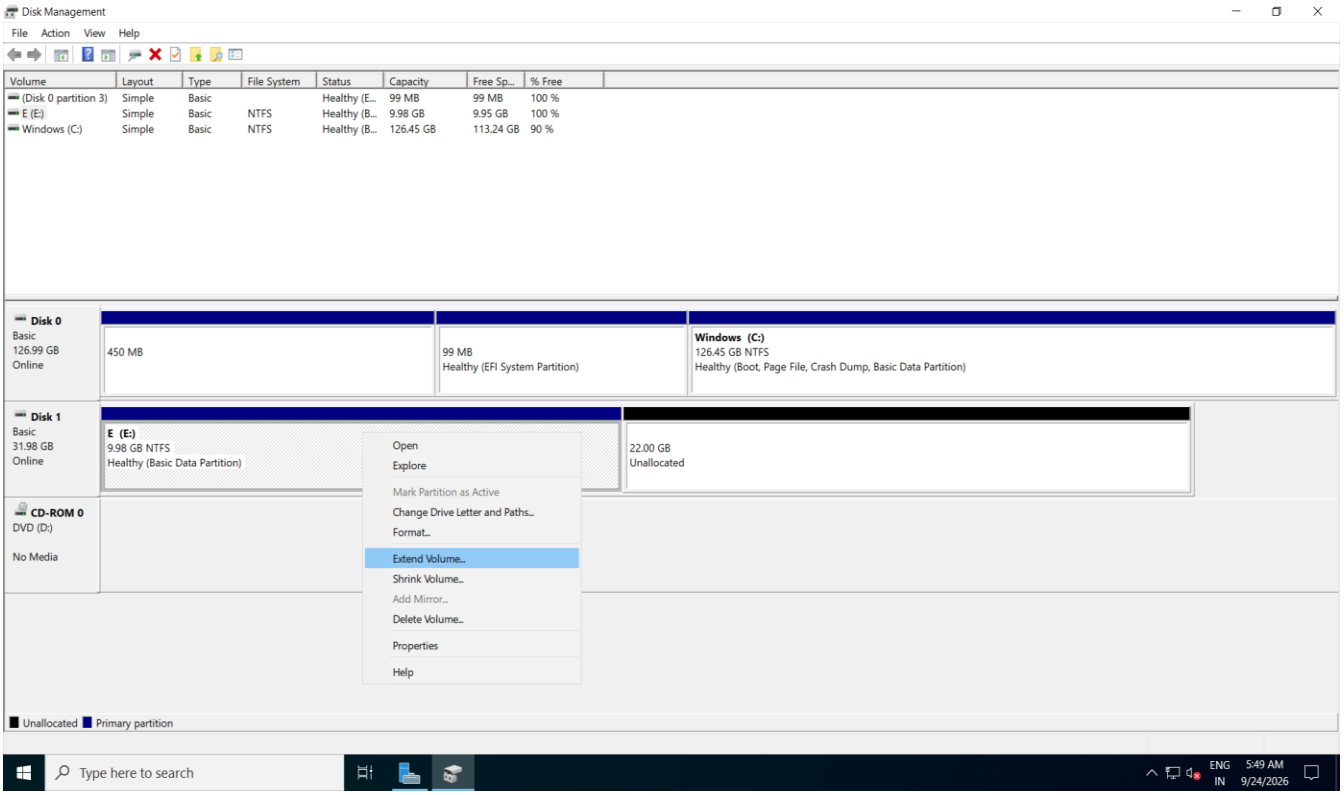
Step 8: Verify Unallocated Space in Windows

1. Connect to the Windows VM using RDP.
2. Open **Disk Management**.
3. Verify the disk now shows **22 GB Unallocated Space**.



Step 9: Extend the Volume

1. Right-click the **E:** drive.
2. Select **Extend Volume**.
3. Complete the Extend Volume Wizard.

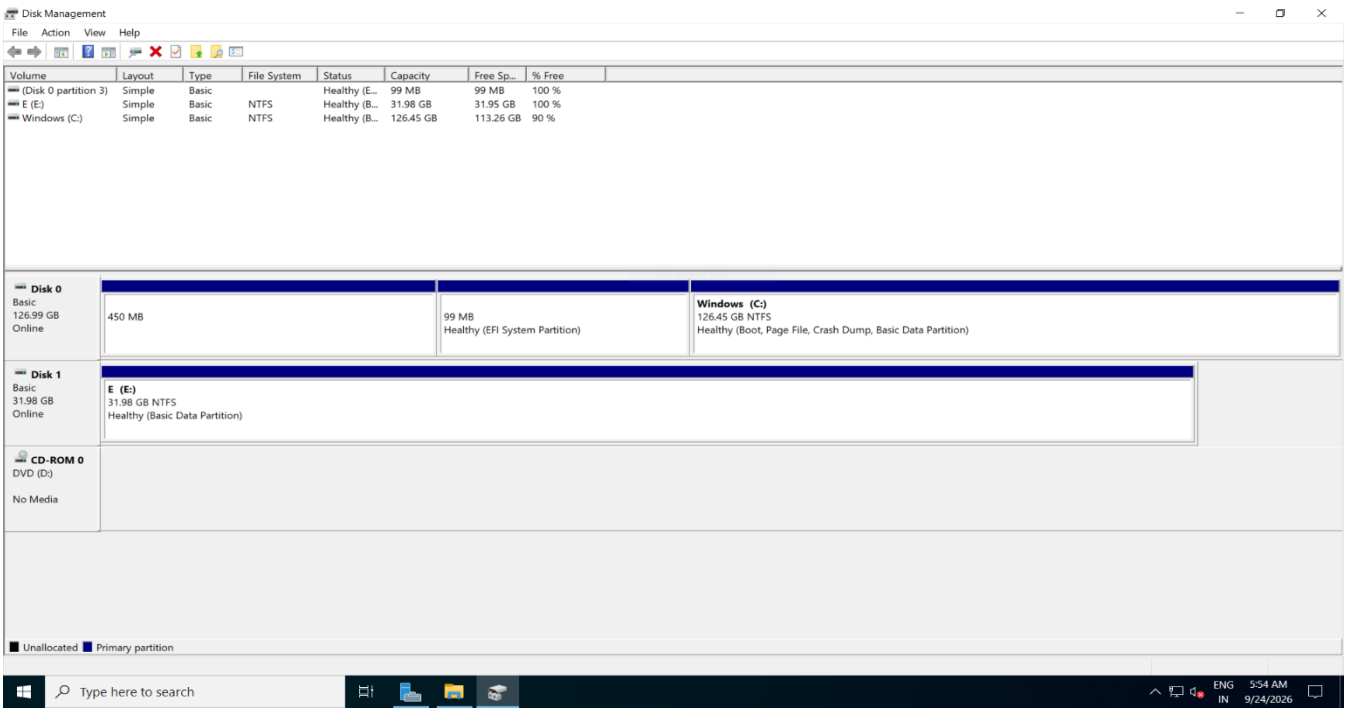


Step 10: Verify Final Disk Size

1. Refresh Disk Management.
2. Verify that the E: drive size has increased successfully.

Final Disk Details

- Drive Letter: E:
- Final Size: 31.98 GB
- Status: Healthy (Basic Data Partition)



Observation

The Azure Managed Data Disk was successfully resized from **10 GiB to 32 GiB** through the Azure Portal. The additional unallocated space was extended within Windows Disk Management and became available for use.

Result

Successfully resized the Azure Managed Data Disk attached to the Windows Virtual Machine from **10 GiB to 32 GiB** and extended the E: volume within the Windows operating system.

Resize the Data Disk of a Linux Virtual Machine

Objective

To increase the size of an Azure Managed Data Disk attached to a Linux Virtual Machine and extend the Linux file system to utilize the newly allocated storage space.

Theory

Azure Managed Disks provide persistent block-level storage for Azure Virtual Machines. When business or application storage requirements increase, an existing managed disk can be resized without recreating the virtual machine.

After increasing the disk size from the Azure Portal, the operating system must detect the new capacity and the file system must be extended so that Linux can use the additional storage space.

Benefits of Disk Resizing

- Increases storage capacity without data loss.
- Eliminates the need to create a new disk.
- Supports growing application requirements.
- Can be performed with minimal downtime.
- Improves storage flexibility and scalability.

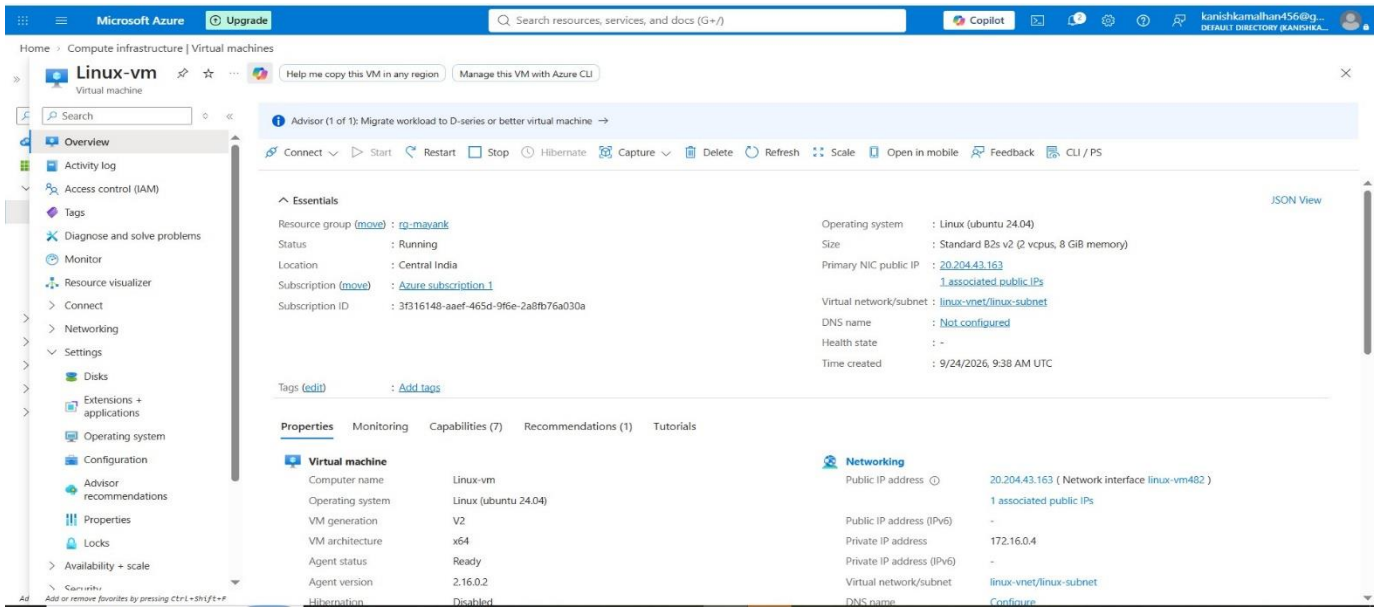
Components Used

- Microsoft Azure Portal
- Azure Linux Virtual Machine (Linux-vm)
- Azure Managed Data Disk (Data_Disk1)
- Ubuntu 24.04 Operating System
- SSH Terminal

Procedure

Step 1: Open the Linux Virtual Machine

1. Sign in to the Azure Portal.
2. Navigate to **Virtual Machines**.
3. Select the virtual machine **Linux-vm**.
4. Verify that the VM is in the **Running** state.

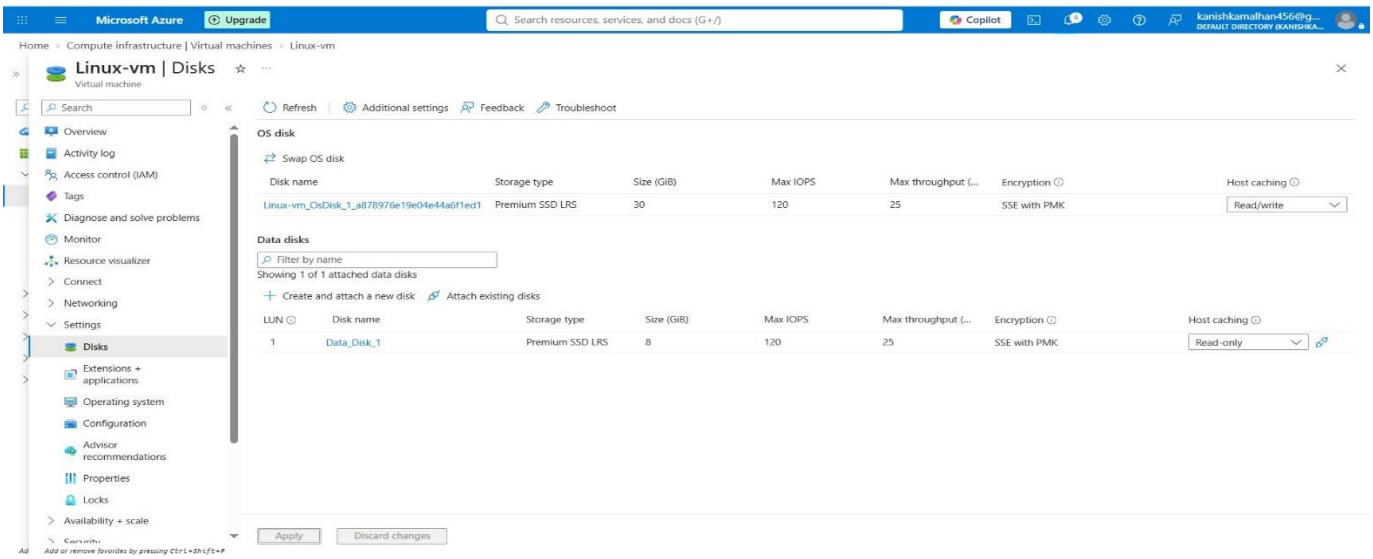


Step 2: Verify Existing Data Disk Size

1. From the left navigation pane, select **Disks**.
2. Review the attached disks.
3. Verify the current size of the data disk.

Existing Disk Details

- Disk Name: Data_Disk1
- Storage Type: Premium SSD LRS
- Disk Size: 8 GiB

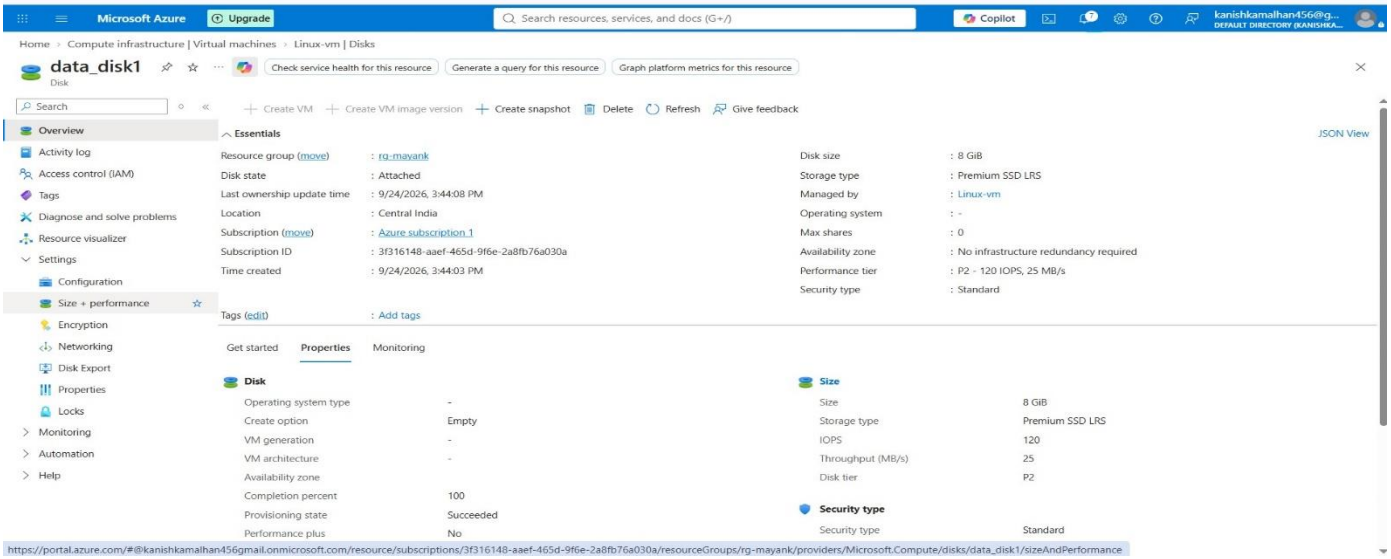


Step 3: Open the Data Disk Overview Page

1. Click on **Data_Disk1**.
2. Review the disk properties.

Disk Information

- Disk State: Attached
- Storage Type: Premium SSD LRS
- Current Size: 8 GiB
- Managed By: Linux-vm
- Performance Tier: P2

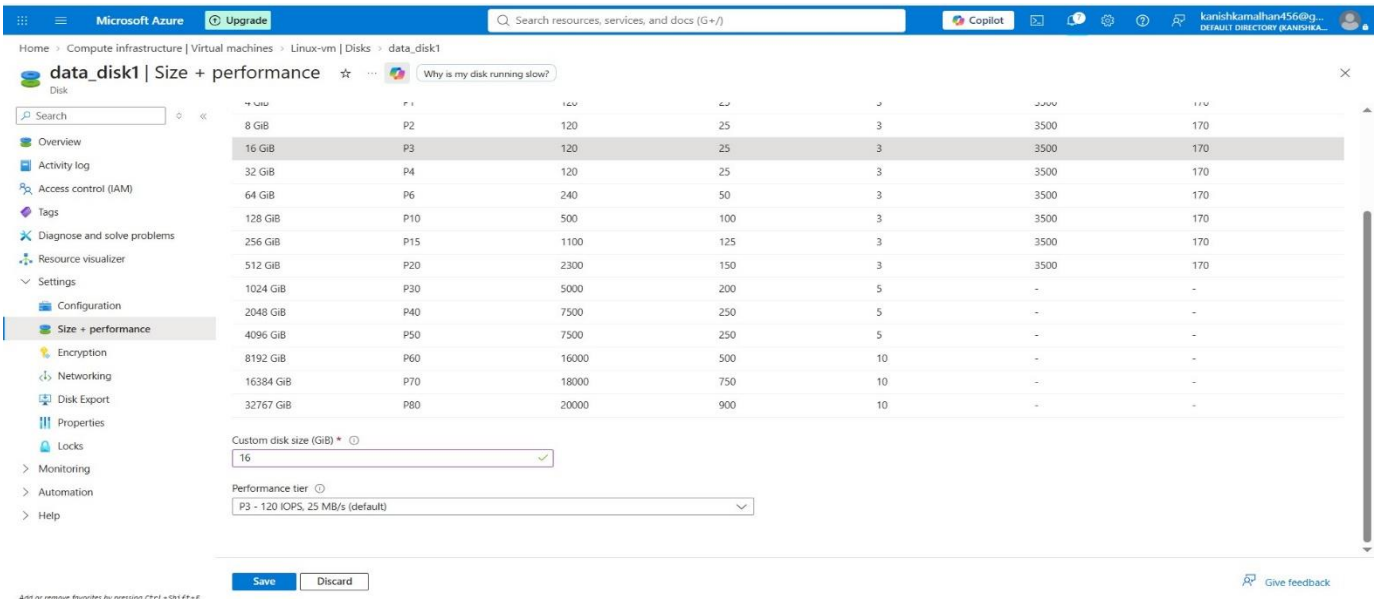


Step 4: Open Size + Performance Settings

1. From the left navigation pane, select **Size + Performance**.
2. Review the available disk size options.
3. Enter the new required disk size.

Resize Configuration

- Existing Size: 8 GiB
- New Size: 16 GiB
- Performance Tier: P3

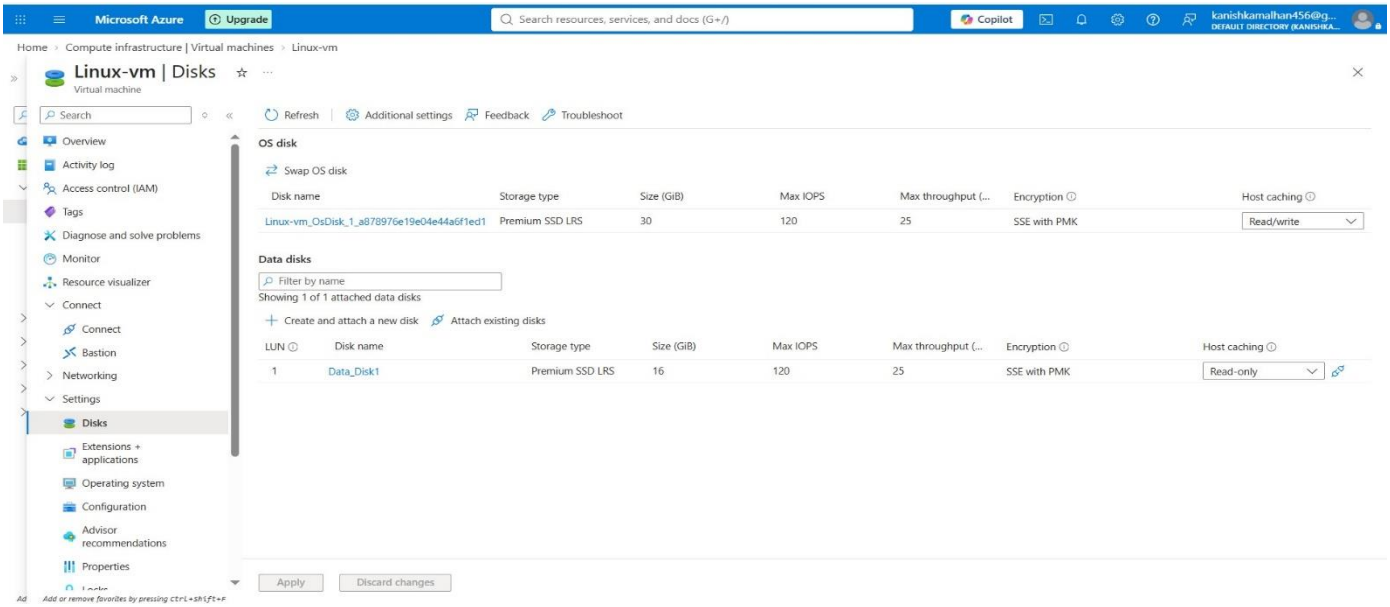


Step 5: Verify Disk Resize in Azure Portal

1. Return to the **VM Disks** page.
2. Verify that the size of **Data_Disk1** has been updated successfully.

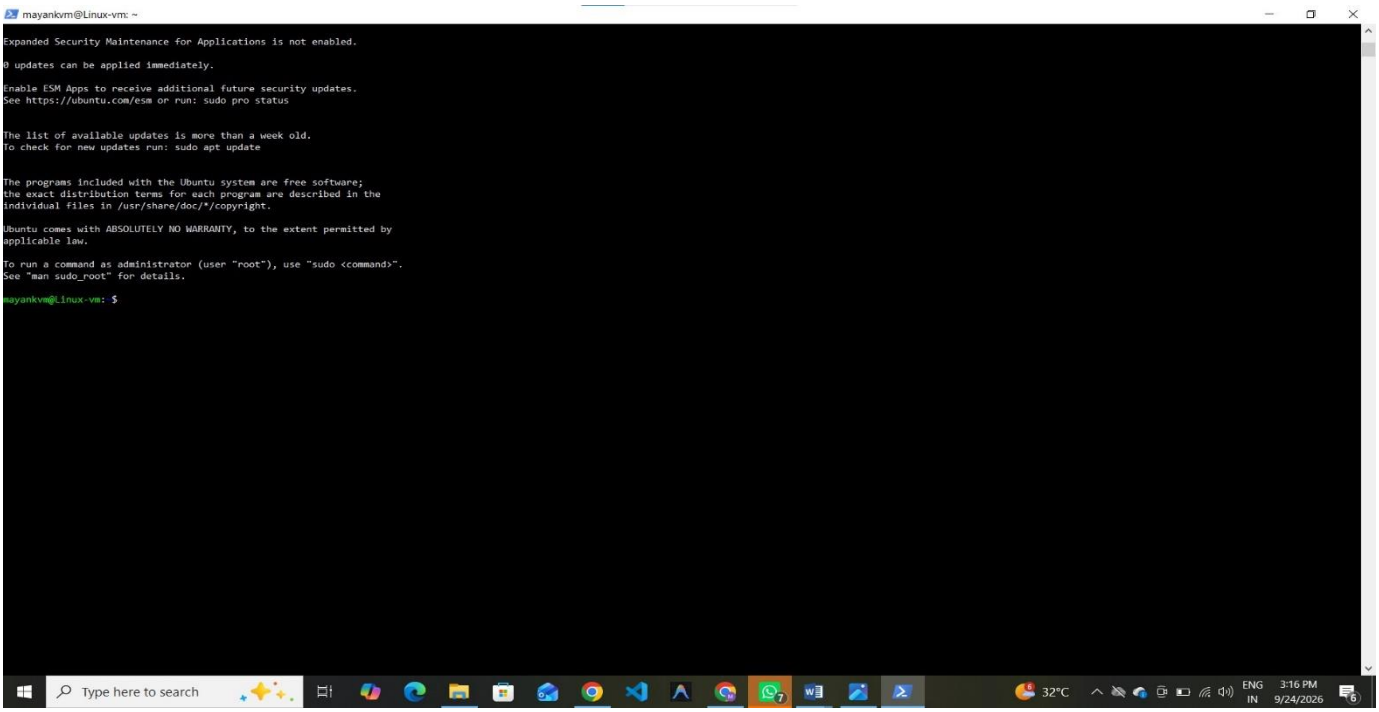
Updated Disk Details

- Disk Name: Data_Disk1
- Storage Type: Premium SSD LRS
- Updated Size: 16 GiB



Step 6: Connect to the Linux Virtual Machine

1. Open the SSH terminal.
2. Connect to the Linux Virtual Machine.
3. Verify successful login.



Step 7: Verify Updated Disk Size in Linux

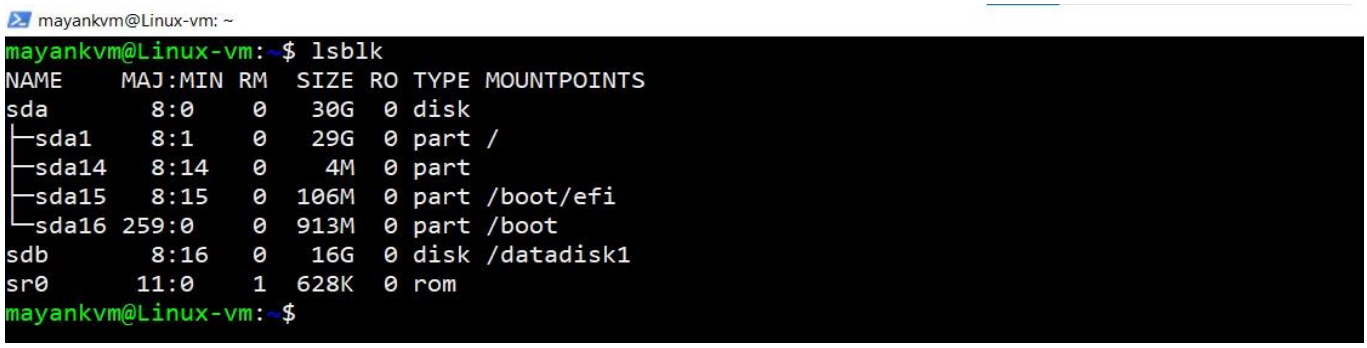
Execute the following command:

- lsblk

Verify that the attached disk size has increased from **8 GiB** to **16 GiB**.

Observation

- OS Disk: sda (30 GiB)
- Data Disk: sdb (16 GiB)
- Mount Point: /datadisk1



Step 8: Extend the Linux File System

Execute the following command:

sudo resize2fs /dev/sdb

The command expands the EXT4 file system and makes the additional storage available to Linux.

Observation

- File system resized successfully.
- Additional disk space allocated to the mounted file system.

```
mayankvm@Linux-vm: ~
mayankvm@Linux-vm:~$ sudo resize2fs /dev/sdb
resize2fs 1.47.0 (5-Feb-2023)
Filesystem at /dev/sdb is mounted on /datadisk1; on-line resizing required
old_desc_blocks = 1, new_desc_blocks = 2
The filesystem on /dev/sdb is now 4194304 (4k) blocks long.

mayankvm@Linux-vm:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        29G  1.7G   27G   6% /
tmpfs           3.9G     0  3.9G   0% /dev/shm
tmpfs           1.6G  996K  1.6G   1% /run
tmpfs           5.0M     0  5.0M   0% /run/lock
efivarfs        128K   38K   86K  31% /sys/firmware/efi/efivars
/dev/sda16      881M   64M  756M   8% /boot
/dev/sda15      105M  6.2M   99M   6% /boot/efi
/dev/sdb         16G   24K   15G   1% /datadisk1
tmpfs           789M   12K  789M   1% /run/user/1000
mayankvm@Linux-vm:~$
```

Step 9: Verify Expanded Storage Capacity

Execute the following command:

- df -h

Verify that the mounted file system now reflects the increased storage capacity.

Final Storage Details

- Disk: /dev/sdb
- Mounted On: /datadisk1
- Total Size: 16 GB
- Available Space: Approximately 15 GB


```
mayankvm@Linux-vm: ~
mayankvm@Linux-vm:~$ sudo resize2fs /dev/sdb
resize2fs 1.47.0 (5-Feb-2023)
Filesystem at /dev/sdb is mounted on /datadisk1; on-line resizing required
old_desc_blocks = 1, new_desc_blocks = 2
The filesystem on /dev/sdb is now 4194304 (4k) blocks long.

mayankvm@Linux-vm:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        29G  1.7G   27G   6% /
tmpfs           3.9G     0  3.9G   0% /dev/shm
tmpfs           1.6G  996K  1.6G   1% /run
tmpfs           5.0M     0  5.0M   0% /run/lock
efivarfs        128K   38K   86K  31% /sys/firmware/efi/efivars
/dev/sda16      881M   64M  756M   8% /boot
/dev/sda15      105M  6.2M   99M   6% /boot/efi
/dev/sdb         16G   24K   15G   1% /datadisk1
tmpfs           789M   12K  789M   1% /run/user/1000
mayankvm@Linux-vm:~$
```

Observation

The Azure Managed Data Disk was successfully resized from **8 GiB to 16 GiB** through the Azure Portal. After verifying the updated disk size within Linux and extending the EXT4 file system using the `resize2fs` command, the newly allocated storage became available for use.

Result

The Azure Managed Data Disk attached to the Linux Virtual Machine was successfully resized from **8 GiB to 16 GiB**, and the Linux file system was extended to utilize the additional storage space.